

Extreme Weather Grid Load Forecasting using Weather-Informed LSTM and Transformer Machine Learning Models

1st Sajib Debnath

School of Computer Sciences
Western Illinois University
Macomb, IL, USA
sajib.uui.cse@gmail.com

2nd Uzzal Basu

School of Engr. & Tech
Western Carolina University
Cullowhee, USA
uzzal.basu@gmail.com

3rd Md Abubakkar

School of Computer Sciences
Midwestern State University
Dallas, TX, USA
mabubakkar@ieee.org

4th Md Robiul Islam

Dept. of Computer Engineering
Tennessee State University
Nashville, TN, USA
rislam1@my.tnstate.edu

5th Mithu Debnath

Wind Performance & Reliability Engineering
AES Clean Energy
Louisville, CO, USA
mithu.debnath@aes.com

6th Ashis Kumer Biswas

Dept. of Computer Science & Engineering
University of Colorado Denver
Denver, CO, USA
ashis.biswas@ucdenver.edu

Abstract—Accurate forecasting of electricity demand has become increasingly challenging due to rising consumption trends, climate variability, and extreme weather events. Historical forecasting models have generally been unsuccessful in capturing nonlinear and sudden transitions in electricity load under these situations. This work proposes a deep learning-based, weather-aware forecasting framework based on LSTM and Transformer architectures to enhance short- and long-term forecasting of electricity demand under extreme weather conditions. Past load and weather data from 2018 to 2024 for Texas are merged and used to train and validate the models with a window of 24 hours. Both LSTM and Transformer models predicted 2024 electricity demand accurately, achieving overall accuracies of 97.8% and 98.6%, respectively. Notably, the Transformer model outperformed the LSTM during periods of extreme demand spikes, attaining an accuracy of 98.7% compared to the LSTM's accuracy of 95.7%. The enhanced performance of the Transformer can be attributed to its global self-attention mechanism, which effectively identifies relationships between meteorological variables and electricity demand, leading to improved prediction of demand surges. These findings underscore the potential of attention-based models to strengthen grid resilience and support climate-resilient energy planning.

Index Terms—LSTM, Transformer, ERCOT, Electricity Load Forecasting, Extreme Weather

I. INTRODUCTION

The US electricity load growth forecast jumps by 81% compared to previous estimates, mostly due to the high demand by the data centers [1]. The growth of artificial intelligence and data centers is demanding more energy from the grid, which now has a wide variety of energy sources, including renewable energy sources. The high demand and large variations of energy sources create congestion in the grid. Electricity demand forecasting can help plan and reduce congestion, ensure proper resource allocation, maintain grid stability, and avoid major disruptions. Major disruptions can occur when

demand and supply are not balanced and proper resources are not allocated ahead of time. Extreme weather phenomena, such as El Niño, the northern polar vortex, cold fronts, and heat fronts, can change the normal patterns of grid demand. If the high demand is not predicted well in these situations of extreme nonstationary and nonlinear cases, such as winter storm Uri in 2021, the grid can fail catastrophically due to improper dispatch planning of the right resources. In Texas, the Electric Reliability Council of Texas (ERCOT) operates one of the largest independent grids in the world, making it a very important case study in terms of load forecasting. Traditional methods of forecasting often fail to capture nonlinear dynamics that exist between extreme weather conditions and electricity consumption, leading to less-than-optimal forecasts. The Autoregressive Integrated Moving Average (ARIMA) is one of the most common methods for time-series forecasting; however, the method struggles with nonlinear data due to its assumption of linear relationships between observed and future series [2], [4]. However, by exploiting long-term dependencies and temporal sequences, LSTM can forecast complex univariate electric load time series with strong non-stationarity and non-seasonality [2], [3]. This study addresses the challenge of load forecasting of electricity in extreme weather conditions by proposing an application of weather-informed machine learning models. During extreme weather events, the spike in electricity demand varies depending on the extremity of the events compared to the recent normal trends and seasons of the year. This study proposes the use of weather-informed machine learning models to address load forecasting challenges during extreme weather events by integrating meteorological variables with historical electricity demand data. We review two state-of-the-art architectures: first, LSTM, a variation of RNN, which is popular in modeling temporal relationships, and

Transformer, which has recently attained popularity due to its efficiency when processing sequential data using self-attention mechanisms. The contributions of this study are as follows:

- Understanding the relationship between weather and electricity demand.
- Integration of sparse weather data into the electricity demand data.
- Development of LSTM and Transformer models for the prediction of electricity demand.
- Testing the performance of the developed models in predicting electricity demand in extreme weather events.

II. DATASETS

The datasets used in this study were composed of two main components: hourly weather data from three different weather stations, publicly available from the National Oceanic and Atmospheric Administration (NOAA) and IOWA Environmental Mesonet [5] and hourly electricity demand data from the ERCOT website [6].

A. Weather Data

Three weather stations (BKS, JDD, and TME) distributed in different regions of Texas, USA are chosen to capture the variations of weather across Texas. The hourly data of air temperature, relative humidity, wind speed, feel-like temperature, and precipitation, spanning from 2018 to 2024, are considered in this study. The missing data points are linearly interpolated to have corresponding values for the electricity demand data.

B. Load Data

The grid load data, from 2018 to 2024, are taken from ERCOT website [6]. The dataset has a measure of electricity demand in megawatts for ERCOT in an hourly manner between 2018 and 2024. The demand data actually capture the total demand for electricity for the entire ERCOT region, representing the aggregated consumption of residential, commercial, and industrial electricity users within the ERCOT region.

C. Data Integration and analysis

Weather data are integrated with ERCOT electricity demand data to analyze the correlation between weather conditions and electricity demand (Fig. 1). This integration was accomplished by aligning the timestamps of the weather data with those of ERCOT load data. Both datasets were averaged hourly to ensure accurate timestamp alignment. The data were pre-processed and analyzed before being injected into the machine learning models. There is a clear relationship between weather variables (air temperature) and electricity demand (Fig. 2). During summer, the electricity demand increases with the increase in temperature (Temperature > 18.5°C), while for the rest of the year, the electricity demand decreases with the increase in air temperature (Temperature < 18.5°C). This is a parabolic curve with two different gradients on two sides with a vertex close to 18.5°C (Fig. 2). During the wind storm Uri in winter 2021, there was a drop in air temperature and a spike

in electricity demand, marked by a vertical black dashed line in Fig. 1, representing an extreme winter event.

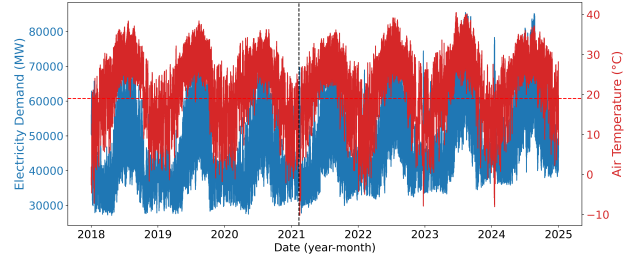


Fig. 1. Timeseries data of electricity demand and air temperature. The vertical black dash line represents the winter storm Uri.

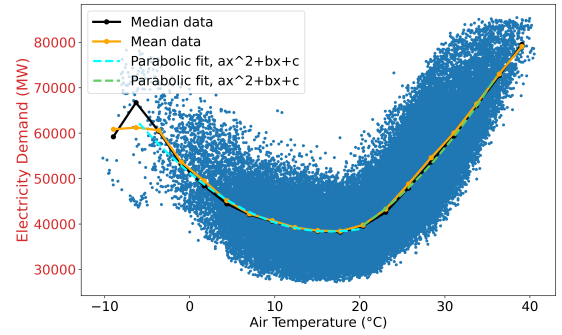


Fig. 2. Relationship between electricity demand and air temperature in Texas.

To model the nonlinear relationship and seasonal dependence of electricity demand on air temperature, [11], [12] we employed a piecewise parabolic function. The relationship is as follows:

$$D(T) = \begin{cases} a_1 T^2 + b_1 T + c_1, & \text{if } T < T_0 \\ a_2 T^2 + b_2 T + c_2, & \text{if } T \geq T_0 \end{cases} \quad (1)$$

Where:

- $D(T)$ is the electricity demand at temperature T
- $T_0 \approx 18.5^\circ\text{C}$ is the vertex temperature (inflection point)
- $a_1 = 47.2, a_2 = 52.4, b_1 = -1560.6, b_2 = -864.5, c_1 = 51230, c_2 = 35523.9$ are coefficients learned via least squares regression.

The curve shows increasing energy demand in cold and warm weather, as anticipated by heating and cooling demands. The first-order derivative of this function, which represents demand sensitivity to air temperature, is as follows:

$$\frac{dD}{dT} = \begin{cases} 2a_1 T + b_1, & \text{for } T < T_0 \\ 2a_2 T + b_2, & \text{for } T \geq T_0 \end{cases} \quad (2)$$

This derivative shows how demand increases more sharply as the temperature moves away from the comfort point. The demand rate depends on the seasons of the year and the air temperature away from the comfort temperature. These steep slopes, especially in severe weather conditions, and dependence on the season of the year, call for the use of deep

learning models capable of learning nonlinear transitions in energy consumption patterns [15].

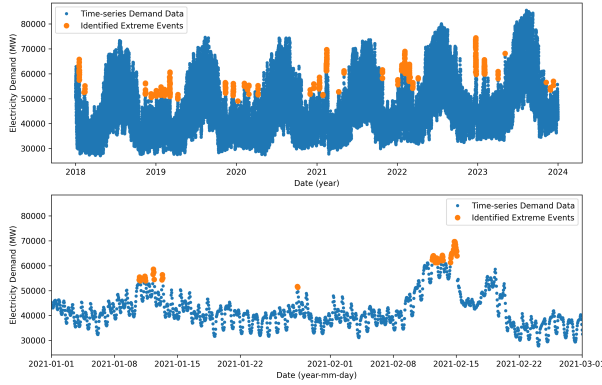


Fig. 3. Extreme energy demand event identification and visualization in the time-series demand data (top); Only two months of data are shown for closer visualization (bottom).

D. Extreme event identification

The extreme energy demand events in this work are defined as the sudden change in demand compared to the trend. The extreme energy demand events from the transient events are identified using the Hampel Filter [26]. The Hampel filter identifies extreme variations by comparing each data point within a sliding window to the median absolute deviation of the window. If a data point deviates from the median by more than a specified number of mean absolute differences, it is flagged as an extreme event. For this analysis, a window of a month and a standard deviation of 3 are used in the Hampel filter to identify the extreme events. The identified extreme events are presented as time series in (Fig. 3). Interestingly, all the identified extreme events occurred in winter, although the summer months had the higher peak demand compared to the winter months. It emphasizes that the demand rates vary a lot in winter months compared to the summer months; thus, the prediction and dispatching of energy are difficult in winter months.

III. MACHINE LEARNING MODELS

The end-to-end process for forecasting electricity demand during extreme weather is shown in Fig. 4, which starts with data collection, where weather and electricity load data are collected. This is followed by a data preprocessing step, where timestamps are aligned, missing values are imputed by linear interpolation, and features are normalized. Later, in the feature integration stage, the preprocessed data are merged into one multivariate time-series split into training and test sets, and sequences are formed using sliding windows. The model training process defines the architectures of the Long Short-Term Memory (LSTM) model, including stacked LSTM layers and dropout, and the Transformer model, including positional encoding, attention feedforward layers, and dense output. The baseline is a Linear Regression model. Adam optimizer is used for training the deep learning models (LSTM and

Transformer), and Mean Squared Error (MSE) is used as the loss function, while Ordinary Least Squares (OLS) is used for the Linear Regression model. Finally, the model performance is evaluated by metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Accuracy, and the results analysis is conducted to identify the effectiveness of the models, especially in predicting extreme events.

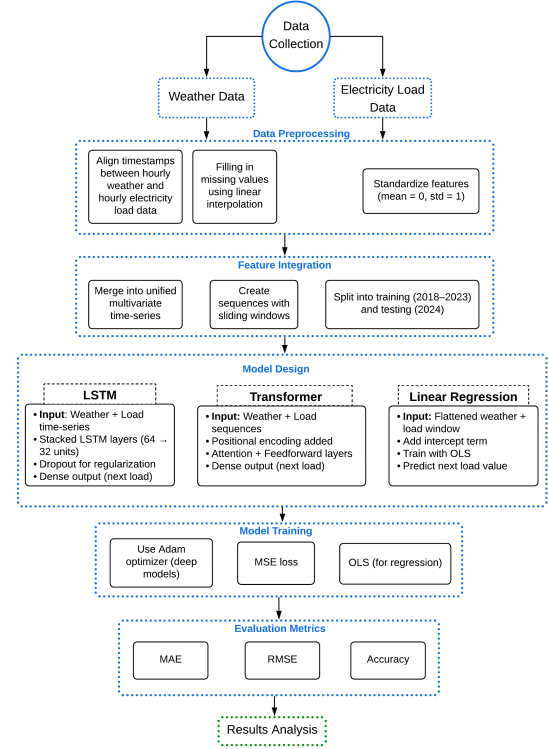


Fig. 4. Workflow Diagram of the Forecasting Process Using LSTM, Transformer, and Linear Regression (Baseline)

This study consists mainly of two advanced machine learning models, Long Short-Term Memory (LSTM) [7] and Transformer, to predict electricity grid loads during extreme weather conditions. Both models are trained with the first 80% of the data from 2018 to 2023. The last 20% of the data of year 2018 to 2023 are used to check the performance of the models (Fig. 7). The machine learning models has no clues about the 2024 data, and the developed model is used to predict the load for the year 2024 based on input features: temperature, wind speed, humidity, and feels-like-temperature. The evaluation of the models is performed on the predicted 2024 load data.

A. Long Short-Term Memory (LSTM) Model

LSTM is an architecture containing multiple layers, the last recurrent layer of which is followed by dense layers. The first layer receives a time series of historical data while each of the LSTM layers learns the [17], [19] temporal dependencies in the data. Dropout layers were set to 20% between layers to prevent overfitting. The last layer is a dense layer with one neuron and predicts the electricity demand for one future time step. The purpose of the model is to explain the nonlinear

interactions between weather variables and electricity demand, particularly regarding extreme weather conditions. LSTM has been designed in such a way that it can model the aforementioned nonlinear relationships quite efficiently; hence it is suitable in the case of extreme weather. The Adam Optimizer highly relies on an adaptive learning rate; hence, it is suitable for large datasets. The mean squared error (MSE) was selected as the loss function. Early stopping was employed to avoid overfitting, and training was performed up to 100 epochs with a batch size of 64. There was a validation set split from the training set for monitoring performances and exploring hyperparameter settings, while generalization to unseen data was guaranteed. The model consists of 80,626 trainable parameters, which indicates its complexity and ability to learn intricate shapes. The training and validation performances are graphically represented using epoch loss in Fig. 5, showing the change in loss and MAE with epochs. The training loss kept on decreasing continuously; this reflected successful learning. The MAE value kept on dropping with time; this indicates continual improvement in the model accuracy. The validation loss (val_loss) mirrored that of the training loss, indicating that the model achieved a fairly good generalization. The validation MAE (val_mae) remained constant, indicating the robustness of the model. The Fig. 5 shows a stable convergence, and for both training and validation metrics, levels off after only a few epochs, which indicates that the model has achieved a balance between bias and variance such that it can be regarded as reliable in the prediction of electricity demand.

The LSTM network is used to represent temporal relationships in electricity load forecasting. In this study, a weather-informed LSTM was used to forecast electricity demand based on historical load and meteorological variables (e.g., temperature, humidity, and wind speed). It consists of a memory cell and three gates—forget, input, and output—to regulate the information flow with time. The forget gate determines what information to forget from before, the input gate determines what information to remember, and the output gate determines what to pass to the next state. [20] The gating mechanism allows the model to hold significant patterns for long sequences [21]. The Fig. 6 shows the processes occurring in the LSTM unit. Each LSTM unit consists of gates that regulate the information flow:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (\text{forget gate}) \quad (3)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (\text{input gate}) \quad (4)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c) \quad (\text{cell input}) \quad (5)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (\text{cell state}) \quad (6)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (\text{output gate}) \quad (7)$$

$$h_t = o_t \odot \tanh(c_t) \quad (\text{hidden state}) \quad (8)$$

Where x_t is the input vector at time t and h_t is the hidden state of the LSTM. [20] The sigmoid activation function is denoted by σ , and the symbol $\odot$ denotes element-wise product. The input connection weight matrices are referred

to as W_f, W_i, W_c, W_o and the recurrent connection weight matrices are referred to as U_f, U_i, U_c, U_o . The bias terms are additionally b_f, b_i, b_c, b_o .

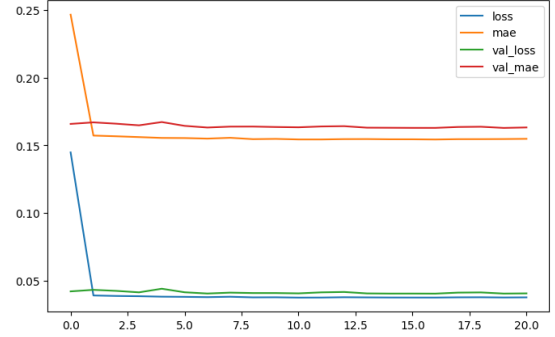


Fig. 5. Training and Validation Loss and MAE over Epochs for LSTM Model.

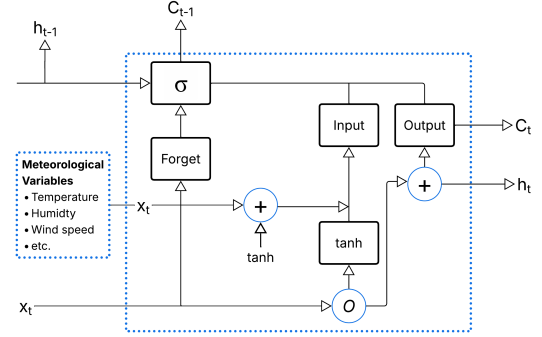


Fig. 6. Weather-Informed LSTM Cell Architecture for Electricity Load Forecasting under Extreme Weather Conditions.

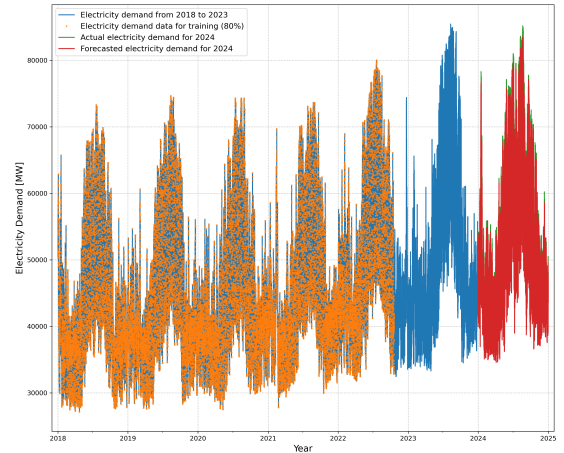


Fig. 7. Training data and forecasting by LSTM for 2024.

B. Transformer Model

The method of applying the transformer model is similar to that of the LSTM model, with some major differences in the architecture of the model and the process of training it

[8]–[10]. The historical electricity demand data and weather data are rescaled according to StandardScaler.

The chosen optimizer for the transformer model was Adam, with the MSE set as the loss function. The training was subjected to a fixed number of epochs, and the training was stopped easily to avoid overfitting. The training dataset was segregated into training and validation sets to monitor the model's performance during training. The validation set was further used for hyperparameter tuning to ensure that the model generalized well to unseen data. Training the Transformer model is computationally demanding owing to its self-attention mechanism, but it is quite effective in capturing long-range dependencies in the field.

The Transformer model is a deep learning architecture built on self-attention, designed to capture long-range dependencies in sequential data without recurrence [8]–[10], [25]. The core operation is scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (9)$$

Where:

- Q , K , and V are the query, key, and value matrices,
- d_k is the dimensionality of the key vectors.

To retain the temporal order, positional encoding was added to the input embeddings using the following functions, as introduced in the original Transformer architecture [25]:

$$\text{PE}_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad (10)$$

$$\text{PE}_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right) \quad (11)$$

The Transformer model used in this study was a time-series electricity grid load forecasting model optimized for severe weather conditions. It accepts input data of shape (batch_size, 24, 13), which is a 24-hour past window of 13 features (electricity demand and weather). [8]–[10] The input was first projected to a fixed size $d_{model} = 64$, and positional encoding was added to maintain temporal information. This is followed by two transformer encoder blocks, each consisting of a multi-head self-attention layer (four heads), a feedforward network with 256 units, dropout regularization, and layer normalization. A global average pooling layer is applied next, along with dropout, and then the last dense layer for predicting the electricity demand. The kind of framework that can abstract deep dependencies in the data and allow for improving forecasting accuracy with nonstationary and abnormal cases. This is a similar principle that is consistent with recent research work in which weather-informed techniques are successful in implementing the optimal power flow in high renewable penetrated transmission networks [24].

C. Linear Regression (LR) Model

In order to create a baseline to compare against performance, a Linear Regression (LR) model was trained using the same dataset with hourly weather data and electricity load. LR

TABLE I
PERFORMANCE COMPARISON OF LSTM, TRANSFORMER, AND LINEAR REGRESSION MODELS ACROSS YEARLY, MONTHLY, AND DAILY DATA WINDOWS.

Model	Data Window	MAE (%)	RMSE (MW)	Acc. (%)
LSTM	Year (2024)	2.2	1489.5	97.8
	Month	2.1	1388.2	97.9
	Day	2.1	1328.9	97.9
Transformer	Year (2024)	1.4	992.4	98.6
	Month	1.4	947.1	98.6
	Day	1.3	925.7	98.7
Linear Regression	Year (2024)	11.1	7538	88.9
	Month	11.3	7212	88.7
	Day	11.2	6687.8	88.8

attempts to find the relationship between the target variable (electricity demand) and a set of input features as a linear combination of the inputs. The general form of multiple linear regression is as follows:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n \quad (12)$$

where $\hat{y}$ is the predicted electricity demand, $x_1, x_2, \dots, x_n$ are the input features (e.g., temperature, humidity, previous load), β_0 is the intercept term, and β_i is the learned coefficient. The model was trained by minimizing the MSE loss function.

In the implementation, each 24-hour window of input data was collapsed into a single vector, eliminating the temporal structure. Consequently, the model treats each input sample in isolation and does not capture the sequential interdependencies. Unlike LSTM and Transformer models, which are specially designed to represent complex, time-varying relationships in the data, the linear regression model cannot learn lagged relationships or react to the nonlinear impacts of extreme weather.

Electricity demand is a transient and non-linear event with a dependency on weather, already been discussed in the Datasets and Methodology section. Previous studies, such as Kamoona et al. [14], used LR models to predict electricity demand. A linear regression model is used in this study as a baseline model to evaluate the performance of nonlinear models. Traditional load forecasting methods, such as linear regression, have shown limitations in dynamic environments, outlined in Taylor and McSharry's evaluations [16].

IV. EVALUATIONS AND METRICS

The performance of the LSTM and Transformer models is evaluated with metrics, namely, Accuracy, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The Accuracy of a model is defined as, $100 - \text{MAE} (\%)$.

Table I presents an in-depth comparison of LSTM, Transformer, and LR models across different data windows, daily, monthly, and yearly. The yearly time-series data are discretized into daily and monthly levels for evaluation on monthly and daily scales. The transformer model performed better than the others across the board on MAE, RMSE, and accuracy

because of its ability to learn long-term dependencies and model complex temporal patterns using self-attention. The LSTM model also performed well, especially in the year and month windows, exposing its strength in handling sequential dependencies. On the other hand, the LR model reflected significantly lower accuracy and higher error rates, with MAE reaching up to 16.0% and RMSE more than 14,000 MW for monthly predictions. These results confirm the inefficacy of linear models in simulating nonlinear and nonstationary electricity demand behaviors, especially under extreme weather conditions [18]. This difference in performance justifies the use of deep learning methods for reliable grid forecasting under extreme conditions and demands.

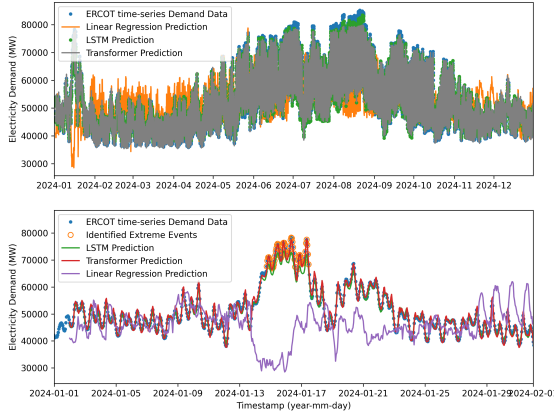


Fig. 8. Forecasting of 2024 electricity demand using LSTM, Transformer, and Linear Regression models (top). Only a month of data with an extreme event is visualized.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (13)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (14)$$

A. Statistical Evaluation of Model Robustness

To assess the reliability of the performance of each model, all experiments were performed in three independent runs with different seed numbers. The performance metrics were averaged, and the standard deviations were checked to determine the variability of the predictions. This approach follows the standard statistical best practices in machine learning model evaluation [23]. For the Transformer model, the MAE was very reliable with multiple runs, yielding $1.4\% \pm 0.10\%$ for yearly forecasting, $1.4\% \pm 0.12\%$ for monthly, and $1.3\% \pm 0.09\%$ for daily forecasts. The LSTM model was less reliable and had MAE's of $2.2\% \pm 0.15\%$ (year), $2.1\% \pm 0.10\%$ (month), and $2.1\% \pm 0.09\%$ (day). The LR baseline model had much larger error values and standard deviations for all the forecasts. A paired t-test was conducted on the daily MAE values for

the Transformer and LSTM model data. It was confirmed that there was a statistically significant difference in model performance with a p-value less than 0.01, which validated the better forecasting accuracy of the transformer model in situations of extreme weather. In conclusion, the transformer model was more accurate, more durable, and more consistent across all benchmarking horizons compared to both the LSTM and LR models.

V. RESULTS

Both the LSTM and Transformer models are trained and evaluated on the same dataset to evaluate their performance in predicting electricity demand under extreme weather conditions. If weather data are incorporated, both models have improved accuracy, demonstrating the importance of meteorological conditions in electricity load forecasting. Across the 2024 horizon, the Transformer and LSTM models have comparable accuracy, MAE, and RMSE. If we focus only on the extreme demand events, there is a considerable difference between the accuracy of LSTM and Transformer models (Fig. 8). A total of six extreme demand events are identified in 2024. The LSTM model achieved an accuracy of 95.7% in forecasting these events, while the Transformer model demonstrated a higher accuracy of 98.7%. Additionally, the RMSE values for the LSTM and Transformer models are 3,261 MW and 1,071 MW, respectively. This suggests that the self-attention mechanism in the transformer is more effective in capturing the complex relationships between weather variables and electricity demand presented in (Fig. 2), enabling the model to forecast demand spikes more accurately.

On the other hand, the LR model, included as a baseline, performed poorly in all horizons. With a mere 80% accuracy and significantly higher error metrics, it failed to detect the nonlinear and time-varying patterns that were present in the data. These results further establish the prowess of deep learning architectures, such as LSTM and Transformer, in modeling dynamic electricity demand, especially under the backdrop of widespread weather variability.

VI. LIMITATIONS AND FUTURE WORK

While the proposed models performed very well, there are certain limitations in this study. Firstly, the models were trained and tested using ERCOT data alone, and generalization to other power grids or sites was not tested. Secondly, while transformer models have been shown to perform better in processing complex dependencies, they are computationally expensive and less suitable for systems with resource-constrained systems and real-time requirements. Third, although weather uncertainties were acknowledged, their direct impact on the precision of the forecast was not measured. Future research could include ensemble techniques for uncertainty estimation, larger regional validation, and real-time adaptation processes for evolving the weather patterns.

VII. DATA AND CODE AVAILABILITY

More details about data preprocessing, feature engineering, and training models can be offered on demand. The original datasets (NOAA and ERCOT) can be accessed through publicly available sources, and the missing weather data [5], [6] are linearly interpolated. The models are written using TensorFlow/Keras with the mentioned hyperparameters. Code and preprocessing scripts are publicly available on release via [GitHub](#) for transparency and reproducibility, aligning with current best practices in machine learning research reproducibility [22].

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